Inclass Exercises – Week 2

Student name: _____

Student ID:_____

(a) Let $x \in \mathbb{Z}$. If x^2 is even, then x is even. If you think that it is true, prove it. Otherwise, give a counterexample.

(b) There exist two distinct positive integers whose sum and difference are perfect squares. Prove it or give an example.

(c) For every integer $n \geq 1$, use proof by induction to show that $5^n - 1$ is divisible by 4.

Additional Exercises:

(a) Use proof by induction on *n* to show that $\sum_{i=0}^{n} x^i = \frac{1-x^{n+1}}{1-x}$ for $x \neq 1$.

(b) Let x and y be two real numbers such that $x, y \geq 0$. Then $\frac{x+y}{2} \geq \sqrt{xy}$. If you think that it is true, prove it. Otherwise, give a counterexample.